



JAIPUR NATIONAL UNIVERSITY

End Semester/Year Examination December -2025

Name of School:	School of Engineering & Technology
Name of Programme / Course:	B.Tech (Computer Science Engineering / AIML / CY / DATA SCIENCE)
Semester/Year:	V / III
Paper Code:	BTECSDS501T22
Title of Paper:	Computer Graphics
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

1. This question paper consists of two sections, Section-A and Section-B.
2. Section-A consists of seven questions and
3. Section-B consists of six questions.

Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question

Question Number	SECTION - A	Marks Allotted	Bloom Level	Course Outcome
Q.1	What are output primitives in computer graphics? List some common primitives and explain their significance.	4	2	1
Q.2	Explain how a point and a line are represented and displayed on a raster display device.	4	3	2
Q.3	Differentiate between Raster Scan and Vector (Random) Scan systems with neat sketches.	4	3	3
Q.4	Describe the Digital Differential Analyzer (DDA) method for drawing a	4	4	4

Q.4

Explain any three techniques used for hidden surface detection in 3D graphics.

10

3

4

Q.5

Describe the Ray Tracing Technique. How are primary rays, reflected rays, and refracted rays generated during the process?

10

4

5

Q.6

Compare RGB, HSV, and CMY color models. Mention where each is commonly used.

10

2

5

line. Why is it considered less efficient compared to Bresenham's algorithm?

Q.5

Use Bresenham's Line Drawing Algorithm to find the pixel positions for a line from (5, 5) to (12, 9).

4

3

5

Q.6

Write short notes on the following:

- (a) Properties of Bezier Curves
- (b) Advantages of B-Spline Curves over Bezier Curves

4

2

5

Q.7

What is 2D Translation? Explain how translation can be represented using homogeneous coordinates.

4

2

1

SECTION - B

Q.1

Explain the Phong Reflection Model. How does it produce more realistic lighting effects?

10

3

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Q.2

Define Computer Animation. Discuss different animation techniques such as keyframe animation, tweening, and motion capture.

10

2

2

Q.3

Write short notes on the following graphical objects:

10

2

3

- (a) Polygon Surfaces
- (b) Parametric Curves
- (c) Quadratic or Quadric Surfaces